

AssessClear AI — 8-Report Evaluation Matrix

Generated directly by local Qwen 3 (14B) on NVIDIA GeForce RTX 2070 SUPER GPU

MODEL ENGINE	COMPUTE ARCHITECTURE	CONTEXT LENGTH	EVALUATION PRIORITY
Qwen 3 (14B - Q4_K_M)	NVIDIA RTX 2070 SUPER (CUDA)	8,192 Tokens	8 Comprehensive Areas (P0-P3)

1. AssessClear Pre-Pilot Research & Findings Report

Key Learnings from Pre-Pilot:

- Strengths Identified:
 - Multi-tiered feedback** (e.g., "What you did well," "Think about this," "Next steps") is pedagogically valuable and easy to understand.
 - Hindi language capability** in Bhopal/Raisen is contextually strong for Madhya Pradesh.
 - Offline scanning** reduces device storage burden (Bhopal/Raisen).
 - Competency-based feedback** shows potential for remedial learning and personalized support.
- Critical Challenges Identified:
 - Dashboard inaccuracies:** Evaluation counts misrepresent actual answer copies assessed (Katni).
 - Session code management:** Codes are not auto-saved, requiring manual note-taking (Katni).
 - AI recognition errors:** Handwriting misinterpretation (e.g., 9996 → 3226), skipped questions, and incorrect MCQ option identification (Katni, Indore).
 - Language and formatting issues:** Hindi PDFs lack accuracy; English feedback is inconsistent (Katni, Indore).
 - Network dependency:** No network in rural areas (e.g., MS Kharbai, Raisen) halts scanning.
 - Dashboard limitations:** 200-evaluation cap, missing subjects/grades, and data loss (Katni, Indore).
 - Time-consuming feedback:** 3–4 minutes per paper for review (Katni).

2. User Experience (UX) & Usability Assessment

Ease of Use and Intuitiveness:

- **Positive Aspects:**
 - **Intuitive navigation** for teachers (e.g., DLs and admins accessing dashboards).
 - **Batch upload mode** simplifies large-scale assessments (Katni).
 - **Clear instructions** for students (e.g., writing sample numbers, not names).
- **Critical UX Issues:**
 - **Dashboard clutter:** Missing subjects (e.g., Hindi in Indore), inconsistent grade visibility, and poor filtering (Indore).
 - **Session management:** No auto-saving of session codes, leading to manual note-taking (Katni).
 - **Feedback complexity:** Time-consuming review (3–4 minutes per paper) and unclear guidance post-scanning (Bhopal/Raisen).
 - **Language barriers:** Hindi PDFs lack accuracy, reducing usability for non-English speakers (Katni).

3. Gap Analysis Report — Expected vs Actual

Expected Feature	Actual Experience
Accurate question assessment	Skipped questions (e.g., Q1–Q2 on first page), incorrect MCQ options (Indore).
Reliable session management	Session codes not auto-saved; manual note-taking required (Katni).
Consistent feedback	PDFs show only teacher feedback later (Katni).
Language accuracy	Hindi PDFs have errors; English feedback is inconsistent (Indore).
Network independence	Network dependency stalls scanning (Raisen).
Dashboard scalability	200-evaluation cap; older data disappears (Katni).
Teacher override capability	No option to edit AI-generated feedback (Indore).

4. Issue & Root-Cause Analysis Report

1. AI Recognition Errors (P0):

Root Cause: Insufficient calibration for complex handwriting (e.g., crossed-out pencil answers, digit misinterpretation).

Impact: Inaccurate scoring (e.g., 9996 → 3226).

2. Dashboard Limitations (P0):

Root Cause: Backend query schema issue (200-evaluation cap).

Impact: Data truncation and incomplete reporting.

3. Session Code Management (P1):

Root Cause: Lack of auto-saving or persistent user profile storage for active session codes.

Impact: Manual note-taking required, increasing friction.

4. Network Dependency (P1):

Root Cause: Lack of offline-first PWA caching and background sync queue.

Impact: Ingestion halts completely in low-connectivity schools (MS Kharbai, Raisen).

5. Language and Formatting Issues (P2):

Root Cause: Inadequate Hindi prompt localization and inconsistent PDF template generation.

Impact: Reduced utility for state Hindi-medium teachers.

6. Feedback Time Consumption (P2):

Root Cause: Overly detailed per-question feedback screens without class-level aggregation.

Impact: High time burden per paper (Katni).

5. Pilot Readiness & Risk Assessment

Readiness Status: Not Fully Ready (High Risk)

- **Blocker Issues:** AI recognition errors, dashboard cap, and zero-network failure must be resolved before full pilot launch.
- **Data Integrity Risks:** 200-evaluation cap truncates historical reporting.
- **Teacher Adoption Risks:** Time-consuming review (3–4 min/paper) risks low teacher engagement.

Mitigation Strategies:

- Prioritize OCR prompt refinement and bounding-box margin fixes.
- Implement offline capture mode with background sync.
- Refactor dashboard queries to support unlimited evaluation history.

6. User Feedback & Thematic Analysis

- **AI Accuracy:** *"AI misinterprets crossed-out answers and skips questions."* (Katni, Indore) · *"9996 → 3226 is a major issue."* (Indore).
- **Dashboard Usability:** *"Dashboard shows 29 evaluations for 3 copies. This is confusing."* (Katni) · *"Missing Hindi and Grade 7/8 in Indore."* (Indore).
- **Language & Formatting:** *"Hindi PDFs are inaccurate. English feedback is inconsistent."* (Katni, Indore).
- **Time and Effort:** *"3–4 minutes per paper for feedback review is impractical."* (Katni) · *"AI should be a tool, not a replacement for teachers."* (Indore).
- **Network Support:** *"No network in Raisen stalls scanning entirely."* (Raisen).

7. Pilot Monitoring & Evaluation Framework

- **Accuracy Rates:** % of correct AI assessments vs. teacher evaluations; % of skipped/misclassified questions.
- **Time Efficiency:** Average time per paper for scanning, feedback review, and batch processing.
- **User Satisfaction:** Teacher and student feedback on usability, accuracy, and Hindi support.
- **Network Resilience:** % of assessments completed offline vs. online; time lost due to network drops.
- **Dashboard Performance:** Data retention beyond 200 evaluations; completeness of subject/grade filters.
- **Feedback Actionability:** % of actionable vs. generic feedback (e.g., classroom pedagogical activities).

8. Recommendations & Product Improvement Roadmap

Priority	Recommendation	Rationale
P0 (Critical)	Fix AI Recognition Errors	Address handwriting misinterpretation (9996 → 3226) and skipped top/bottom questions.
P0 (Critical)	Expand Dashboard Capacity	Remove 200-evaluation cap; ensure full historical data retention.
P0 (Critical)	Implement Offline Mode	Enable scanning and local queueing in low-network areas (Raisen).
P1 (High)	Auto-Save Session Codes	Store session codes in user profile for instant 1-tap retrieval.
P1 (High)	Improve Hindi Language Accuracy	Localize PDF feedback text for Madhya Pradesh state context.
P2 (Medium)	Add Teacher Override Functionality	Allow teachers to manually correct AI-generated grades/scores.
P2 (Medium)	Simplify Feedback Review	Develop class-level summary screens to reduce review time per paper.
P3 (Low)	Enhance Dashboard Filtering	Add granular filters for subjects, grades, and competencies.
P3 (Low)	Expand Test Scenarios	Pilot with 30–40+ students and open-ended descriptive questions.